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QUICK CONNECT BALL COUPLER

Abstract

A self-latching ball coupler has a lever that rotates on an axis between a latched position and a released position. In the latched position, the coupler can receive and latch to a hitch ball that is moved into a ball pocket or retain the hitch ball that is located in the ball pocket. A spring is in biased contact with the lever that urges the lever towards the latched position when it is away from the released position. In the released position, the spring is in biased contact with a different portion of the lever to retain the lever in the released position. Once the lever is in the released position and the hitch ball is moved out of its ball pocket, the hitch ball contacts the lever, which moves it away from the released position and the spring urges the lever to the latched position.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation-in-Part of U.S. application Ser. No. 18/766,933 filed Jul. 9, 2024, which is a Continuation-in-Part of U.S. application Ser. No. 18/318,863 filed May 17, 2023, which is a Continuation-in-Part of U.S. application Ser. No. 17/888,663, filed Aug. 16, 2022, the disclosures of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

[0002] The present disclosure relates to couplers between a towing vehicle and a trailer. The most common coupler involves a ball and socket, which provides a secure connection that still allows freedom of movement between the vehicle and trailer. Ball and socket couplers have been around for many years and have proven to be a reliable and flexible connection, but can be difficult to align, mate, and secure. Others have attempted to design a simple coupler, but these can still require significant hand strength to operate, have dangerous pinch points, or be difficult to release. Other devices, such as the device shown in U.S. Pat. No. 6,113,125, allow for easy coupling of a hitch but this device must be held in a release position to release the hitch ball. There is no automatic retention in the release position or automatic reset of the coupler for receiving the hitch ball once removed. For these reasons, an improved ball coupler is needed.

SUMMARY OF THE INVENTION

[0003] The present disclosure describes a self-latching ball coupler for coupling and releasing a hitch ball having a diameter. The ball coupler has a housing with a fixed ball pocket complementary to the diameter of the hitch ball. The fixed ball pocket has an opening and the housing has a slot adjoining and extending into the fixed ball pocket. The housing has a lever shroud portion flanking the slot that extends beyond the opening in a direction opposite the fixed ball pocket. The coupler has a lever located in the slot and that is rotatable about a shaft axis between a latched position and a released position. The lever has a concave ball engagement surface spaced from the shaft axis by a first distance, and the concave ball engagement surface extends to a concave ball clearance cove surface that is spaced from the shaft axis by a second distance. The concave ball clearance cove surface terminates at a reset portion. In the latched position, the concave ball engagement surface faces the fixed ball pocket. In the released position, the concave ball clearance cove surface faces the fixed ball pocket. The coupler includes a release handle fixed with respect to the lever and is locked in rotation therewith. The coupler includes a spring in biased contact with the lever. When the lever is in the released position, only the spring retains the lever in the released position. When the lever is moved away from the released position, the spring urges the lever towards the latched position. The hitch ball is removable from the fixed ball pocket when the lever is in the released position. When the lever is in the released position, the reset portion extends into the opening to narrow it to be smaller than the diameter of the hitch ball, and when the hitch ball is removed from the fixed ball pocket, the hitch ball contacts the reset portion and moves the lever away from its released position.

[0004] In addition, the spring has a fixed portion in biased contact with the housing and a moving portion connected to the fixed portion. The moving portion has a follower portion and the lever has a cam surface. The follower portion is spaced from the cam surface when the lever is in its released position. The follower portion is in biased contact with the cam surface when the lever is moved away from its released position. When the lever is in the latched position and the hitch ball is

moved into the fixed ball pocket, the lever remains in biased contact with the hitch ball and the follower portion remains in contact with the cam surface. The spring is formed from a strip of flat material and the follower portion is obliquely angled with respect to the moving portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a top isometric view of a coupler in use connecting a trailer and towing vehicle;
[0006] FIG. 2 is a top isometric view of the coupler in FIG. 1;
[0007] FIG. 3 is a bottom isometric view of the coupler in FIG. 2;
[0008] FIG. 4 is an exploded isometric view of the coupler in FIG. 2;
[0009] FIG. 5 is an exploded isometric view of the housing alignment to the gooseneck tube and ball pocket;
[0010] FIG. 6 is a side section view 6-6 of the coupler in FIG. 2 in the captured position;
[0011] FIG. 7 is a side section view 6-6 of the coupler in FIG. 2 in the released position;
[0012] FIG. 8 is a top section view 8-8 of the coupler in FIG. 2 in the captured position;
[0013] FIG. 9 is a top isometric view showing a bumper-mounted embodiment of the hitch coupler in use connecting a trailer and towing vehicle;
[0014] FIG. 10 is a top isometric view of the coupler in FIG. 9;
[0015] FIG. 11 is a top isometric view of a manually-operated embodiment of the coupler shown in FIG. 1;
[0016] FIG. 12 is a bottom isometric view of the coupler shown in FIG. 11;
[0017] FIG. 13 is a side section view 13-13 of the coupler in FIG. 11 in the captured position;
[0018] FIG. 14 is an exploded isometric view of the coupler in FIG. 11;
[0019] FIG. 15 is a top section view 15-15 of the coupler in FIG. 11;
[0020] FIG. 16 is a top section view 15-15 of the coupler in FIG. 11 after being moved to the unlocked position;
[0021] FIG. 17 is a front isometric view of a self-resetting embodiment of the coupler;
[0022] FIG. 18 is an exploded view of the coupler in FIG. 17;
[0023] FIG. 19 is a rear view of the coupler in FIG. 18;
[0024] FIG. 20 is an exploded view of the locking mechanism of the coupler in FIGS. 17 and 18
[0025] FIG. 21A is an isometric view of the locking mechanism components in FIG. 20 in the locked/captured position;
[0026] FIG. 21B is a section view 17-17 of the coupler in the locked/captured position;
[0027] FIG. 22A is an isometric view of the locking mechanism components in FIG. 20 in the reset position;
[0028] FIG. 22B is a section view 17-17 of the coupler in the reset position;
[0029] FIG. 23A is an isometric view of the locking mechanism components in FIG. 20 in the release position;
[0030] FIG. 23B is a section view 17-17 of the coupler in the release position;
[0031] FIG. 24 is an isometric view of an alternate gooseneck embodiment of the self-resetting coupler;
[0032] FIG. 25 is a section view 25-25 of the gooseneck coupler in FIG. 24;
[0033] FIG. 26 is a top front isometric view of an a-frame and flat mount trailer coupler;
[0034] FIG. 27 is a partially exploded isometric view of the a-frame trailer coupler in FIG. 26;
[0035] FIG. 28 is a bottom rear isometric partially exploded view of the coupler in FIG. 27;
[0036] FIG. 29 is a section view 29-29 of the coupler in FIG. 26;
[0037] FIG. 30 is an isometric view of the locking pin and related components;
[0038] FIG. 31 is a top front isometric view of an alternate mounting of the coupler in FIG. 26;

[0039] FIG. **32** is a topfront isometric view of an alternate embodiment of the coupler;
[0040] FIG. **33** is an exploded isometric view of the coupler in FIG. **32**;
[0041] FIG. **34** is a bottom rear exploded isometric view of the coupler in FIG. **33**;
[0042] FIG. **35** is a bottom view of the coupler in FIG. **32**;
[0043] FIG. **36** is a side section view **36-36** of the coupler in FIG. **35**;
[0044] FIG. **37** is a bottom isometric view of an alternate embodiment of the coupler;
[0045] FIG. **38** is an exploded isometric view of the coupler in FIG. **37**;
[0046] FIG. **39** is a front section view **39-39** of the coupler in FIG. **37**;
[0047] FIG. **40A** is a side section view **40-40** of the coupler in FIG. **37** as a hitch ball is entering the fixed ball pocket with the lever in the latched position;
[0048] FIG. **40B** is a side section view **40-40** of the coupler in FIG. **37** as the hitch ball has rotationally displaced the lever away from the latched position;
[0049] FIG. **40C** is a side section view **40-40** of the coupler in FIG. **37** with the hitch ball secured within the fixed ball pocket and the lever in the latched position;
[0050] FIG. **40D** is a side section view **40-40** of the coupler in FIG. **37** with the lever moved to the released position; and
[0051] FIG. **40E** is a side section view **40-40** of the coupler in FIG. **37** with the hitch ball moving the lever from the released position as it exits the coupler.

DESCRIPTION OF THE PREFERRED EMBODIMENT

[0052] A coupler **10** is designed to mate to a standard hitch ball **12** having a partially spherical surface **14**. The hitch ball **12** is typically affixed to a towing vehicle **16**, either at the rear of the vehicle as shown in FIG. **9** or above an axle as shown in FIG. **1**. When the hitch ball **12** is affixed to the vehicle frame above an axle, it is typically when the vehicle is set up for towing a gooseneck trailer. As is common with hitch balls, the hitch ball **12** contemplated herein has a spherical portion and a stem portion, with the stem portion being smaller than the spherical portion. The couplers discussed herein capture the hitch ball **12** by trapping the spherical portion while still allowing rotation with respect to the coupler.

[0053] The coupler **10** shown in FIGS. **1-8** is useable with a gooseneck trailer, with a manual coupler **110** shown in FIGS. **11-16**, and a bumper version of the coupler **210** shown in FIGS. **9-10** being useable with a bumper-mounted or rear-mounted hitch ball. For the purposes of simplicity, the gooseneck version of the coupler **10** will be described in detail. The bumper version of the coupler **210** will then be described to the extent that it differs from the coupler **10**. In the disclosures herein, the couplers **10**, **110**, **210** use the same operative parts, with some structural differences being related to how the locking pin is released and/or how the coupler mounts to the trailer (not shown). The coupler **10**, **110**, **210** is mounted to a trailer with a coupling axis **50** perpendicular or substantially perpendicular to the ground. Because vehicles and/or trailers can be located on unlevel ground, substantially perpendicular is defined as the ball pocket being sufficiently vertical for the coupler to mate with the hitch ball **12**.

[0054] The coupler **10** has a fixed ball pocket **20** with a cylindrical inside surface **22** extending to a hemispherical surface **24**. The cylindrical inside surface **22** and hemispherical surface **24** cooperate to form a partial capsule-shaped inside envelope **28**. In other words, the surfaces **22**, **24** combine to form one end of a spherocylindrically shaped cavity. Other names for this shape include a hemisphere cylinder. The fixed ball pocket **20** has a notch **26** that interrupts the cylindrical inside surface **22** and extends into the hemispherical surface **24**. A lever **30** is retained in the notch **26** that is used to capture the hitch ball **12**. The lever **30** pivots between a captured position (shown in FIG. **6**) and a released position (shown in FIG. **7**) about a pin **32** that is secured to a housing **40**. The pin **32** is shown as being held in with a clip **33**, but other methods of securing the pin **32** are contemplated. The housing **40** is fixed with respect to the fixed ball pocket **20**, typically through welding. For coupler **10**, the fixed ball pocket **20** and housing **40** are secured to a gooseneck tube **21** that is made to be secured to a trailer **18**, as shown in FIG. **1**. The fixed ball pocket **20** has

grooves **23** that may be used to align with features on the housing **40**, which can ensure alignment and proper positioning of the lever **30** with respect to the fixed ball pocket **20**. The lever **30** is biased towards the released position through gravity, because the aperture for the pin **32** is offset from the center of gravity of the lever. It is contemplated that the lever **30** is biased with a spring or other component. The lever **30** has a locking aperture **34** that aligns with rack apertures **42** in the housing **40** when the lever is in the captured position. In the captured position, the lever **30** also has a manual locking aperture **36** that aligns with manual lock apertures **44** in the housing **40**. The manual lock apertures **44** allow for a keyed or other security lock **48** to be installed. The lever **30** has a ball engagement surface **38** that is complementary to the partially spherical surface **14** of the hitch ball **12**. The lever **30**, and in particular, the ball engagement surface **38** portion extends into the inside envelope **28** irrespective of the lever's position. An upper portion of the engagement surface **38** extends inwardly when the lever **30** is in the release position and a lower portion of the engagement surface extends inwardly in the captured position. This is shown in FIGS. **6** and **7**. At points between these two extremes, one portion or both portions extend into the inside envelope **28**. [0055] As previously disclosed, the lever **30** and housing **40** have apertures that align when the lever is in the captured position. The housing **40** includes a slidable locking pin **60** that moves along a locking axis **62** between a locked and unlocked position. The slidable locking pin **60** has a rack gear **64** and a spring hook **66**. The spring hook **66** is attached to a spring **70** that is connected to the housing **40** at a spring eyelet **46**. The spring **70** biases the locking pin **60** towards the locked position, which is shown in FIG. **2**.

[0056] The rack gear **64** mates with a spur gear **80** that is affixed to an actuator or motor **82**. The motor **82** can rotate the spur gear **80** to move the locking pin **60** along the locking axis **62**. The motor **82** is mounted to a bracket **84** and is electrically connected to a control **90**. While the motor **82** and control **90** are shown as separate components, it is contemplated that the control and motor are integrated into a single unit or enclosure. The control **90** houses electronics that are either battery-powered, powered from the trailer, or powered by the towing vehicle. Optionally, a sensor is located in the fixed ball pocket **20** or another location to detect the presence of the hitch ball **12**. The sensor can be magnetic, proximity, or any type of sensing technology that detects the presence or movement of the hitch ball, lever, or other components discussed herein. When the hitch ball **12** seats into the fixed ball pocket **20** or is removed from it, the sensor sends a signal to the control **90**. It is contemplated that the sensor is located to detect the position of the lever **30**. The sensor is in electrical communication with the control **90** and can be used to notify the user of the status of the hitch and/or be used to move the motor **82**. The electronics control the motor **82** and can receive signals from a wireless remote, communicate over Bluetooth, NFC, or other protocol. Bluetooth, NFC, or other protocols are used with smartphones, tablets, or other mobile devices that would have a custom program/application. The application would provide the status of the motor, sensor(s), and battery level, along with providing control for the motor to release the coupler.

[0057] The manual version of the coupler **110** is shown in FIGS. **11-16**, where the motor, controller, and optional sensor are not present. Coupler **110**, like coupler **10**, uses the fixed ball pocket **20** and lever **30**. The differences between coupler **10** and coupler **110** are related to how the locking pin **160** is moved to the unlocked position. A first housing side **140** and a second housing side **141** cooperate with a guide bracket **184** to form a housing. The first and second housing sides **140**, **141** have apertures **142** for the locking pin **160**. The locking pin **160** slides along a locking axis **162**, shown in FIG. **14**. As shown in FIGS. **15-16**, the locking pin **160** is attached to a release lever **182** and biased towards the locked position by a spring **170**. The spring **170** is held between a guide bracket **184** and a step **166** located on the locking pin. The step **166** could be integrated into the locking pin **160** or be a clip (such as a c-clip or e-clip) attached thereto. The second housing side **141** has a pocket **143** to receive the step **166** in the locked position. The locking pin **160** has a guide flat **164** that cooperates with an aperture **183** in the guide bracket **184** to prevent the locking pin **160** from rotating and maintain alignment and orientation of the pin and release lever. The

release lever **182** is tied to the locking pin **160** by a lever pivot **186** which allows the release lever **182** to pivot with respect to the locking pin **160**. The release lever **182** has a rounded surface **188** that slides along the guide bracket **184** and an end surface **190** that can hold the locking pin **160** in the unlocked position.

[0058] The bumper version of the coupler **210** is shown in FIGS. **9-10** and may either be manual, like coupler **110** or electrically-actuated like coupler **10**.

[0059] As previously described, the coupler **10**, **110**, **210** releases and captures the hitch ball **12** through the movement of the lever **30**. When the hitch ball **12** is absent from the inside envelope **28** and the locking pin **60**, **160** is in the unlocked position, the lever **30** is free to move between the release and capture positions but will naturally rotate to the release position through gravity. This puts the coupler **10**, **110**, **210** into a condition to receive the hitch ball **12**. The upper portion of the engagement surface **38** extends into the inside envelope **28** in this position. At the same time, the locking pin **60**, **160** is biased towards the locked position but cannot reach it because the locking aperture **34** is not aligned. Instead, the locking pin is biased against the side of the lever **30**. When the hitch ball **12** is moved into the inside envelope **28**, it meets the upper portion of the engagement surface **38**, where it begins to rotate the lever **30** towards the captured position. As the hitch ball **12** reaches a fully seated position against the hemispherical surface **24**, the lever **30** reaches the captured position, which aligns the locking aperture **34** with the locking pin **60**, **160**. The bias pressure on the locking pin (from the spring **70**, **170**) causes it to extend through the locking aperture **34** completely and through apertures **42**, **142**. The hitch ball **12** is held in place by the lower portion of the engagement surface **38** while the lever **30** is locked from rotating by the locking pin. To release the hitch ball **12**, the user pulls the locking pin **60**, **160** out of the locking aperture **34**. For the manual versions, this is done by actuating the handle **182**, while in the electrically-actuated version it is moved by the motor **82**. Removing the locking pin from the locking aperture allows the lever **30** to swing to the released position, thereby allowing the hitch ball **12** to withdraw.

[0060] Coupler **310** is shown in FIGS. **17-19** and shares many of the same features as couplers **10**, **110**, **210** with a fixed ball pocket **322** having a hemispherical surface **324** on a housing **340**. The fixed ball pocket **322** has an opening **326**. As with the other couplers, coupler **310** includes a lever **330** that pivots between a captured position (shown in FIG. **21B**) and a released position (shown in FIG. **23B**) about a pin **32**. Located between the captured and release position is a reset position, shown in FIG. **22B**. A release handle **382** is attached to the locking pin **360**, which allows it to rotate around and slide along a locking axis **362**.

[0061] The locking pin **360** contains additional features, such as a pair of modified helical grooves **364**. As shown in FIG. **20**, the grooves **364** are located on a minor diameter **366** of the locking pin. The grooves **364** begin on an open end **361** of the locking pin and are diametrically opposed from each other. The grooves **364** spiral around and terminate at an end **368**. Adjacent the end **368** is a knee **372** where the groove changes direction and turn back towards the end of the locking pin. In other words, the knee **372** is a slope change or tipping point along the length of the pin. As will be described later, the knee **372** represents a point where the locking pin **360** moves to either a locked position or an unlocked position. The locking pin **360** has a minor diameter **373**, a major diameter **374**, and a shoulder **375** therebetween. A trip cam **376** is located on a terminal end of the major diameter. The trip cam **376** is shown in FIG. **21** where a biasing surface **378** and a trip surface **380** are located. As will be described later, the trip cam **376** mates with surfaces and features that form a trip notch **336** on the lever **330**.

[0062] Affixed to the housing **340** is a guide collar **390** with a central aperture **392**. Protruding into the central aperture are guide posts **394**. The guide posts **394** are diametrically opposed and are sized to accommodate the grooves **364**. The central aperture is sized to allow the minor diameter **366** to slide through while the guide posts **394** cooperate with the grooves **364** to rotate the locking pin **360** on its locking axis **362**.

[0063] A spring **370**, shown in FIG. **18**, is located over the minor diameter **366**, which provides the biasing force to move the pin to the locked, armed, and release positions. For clarity, the spring **370** is not shown in all figures.

[0064] The lever **330** includes the trip notch **336** that cooperates with the trip cam **376**. The trip notch **336** has a resetting surface **350** and a biased surface **352**. The lever **330** rotates between a captured position (FIG. **21**), a released position (FIG. **22**), and a reset position (FIG. **23**). In the captured position, the locking pin **360** overlays a locking surface **334**, which prevents the lever **330** from moving away from the captured position. In the reset position, the locking pin **360** is clear of the lever **330**, which allows it to rotate about the pin **32**. However, the lever **330** in the reset position does not rotate enough to fully clear the hitch ball **12** to allow it to be inserted or removed. In the release position, the hitch ball **12** can be inserted or removed. The lever **330** includes an engagement surface **338** which contacts the hitch ball **12**. The engagement surface **338** has an upper portion **342** and a lower portion **344**. The upper and lower portions **342**, **344** engage with the hitch ball **12** in different ways and at different times, based on the position of the lever **330** and engagement position of the hitch ball **12**. For example, in the locked position shown in FIG. **21B**, the engagement surface **338** is aligned with the outside surface of the spherical portion of the hitch ball and the lower portion **344** extends partially across the opening **326** to capture the spherical portion of the hitch ball **12**. In other words, the distance across the opening in the locked position is smaller than the spherical portion of the hitch ball, as shown in FIG. **21B**. In the release and reset positions, the upper portion **342** extends into the hemispherical surface **324** of the fixed ball pocket **322**, shown in FIGS. **22B** and **23B**.

[0065] In cooperation with the lever, the locking pin **360** is moveable between a locked position (FIG. **21A**), an armed position (FIG. **22A**), and a released position (FIG. **23A**). In all of these positions, the guide posts **394** are located in different areas of the grooves **364**. This is shown in FIGS. **21-23**. In the released position, the guide posts **394** are located in the end **368**. In the locked position, the guide posts are at the terminal end of the grooves **364** adjacent the release handle **382**. In the armed position, the guide posts **394** are located adjacent the knee **372** and on the opposite side of the knee from the end **368**. In other words, in the armed position, the guide posts **394** are located between the knee **372** and the open end **361**.

[0066] To move the coupler **310** to receive or release a hitch ball, the user first moves the locking pin **360** to the released position by pulling and rotating the release handle **382** until it is in the released position. When the locking pin is in the released position, the guide posts **394** are located in the ends **368**. As the hitch ball **12** is installed or removed, the lever **330** rotates towards the reset position, which causes the resetting surface **350** to contact the trip surface **380**. As the hitch ball **12** continues to rotate the lever **330**, the resetting surface **350** continues to contact the trip surface **380**, which causes the locking pin **360** to rotate and move the guide posts **394** from the ends **368** and over the knee **372** to the armed position. At this point, the locking pin **360** is biased against the lever **330** with the biasing surface **378** in biased contact with the biased surface **352**. As the hitch ball moves into the housing and the lever **330** rotates to the captured position, the locking surface **334** clears the locking pin **360**, allowing the pin to freely slide to the locked position.

[0067] Coupler **410** is shown in FIGS. **24-25** and contains many of the same features and components as coupler **310**. For example, coupler **410**, has a fixed ball pocket **422**, housing **440**, lever **430**, and release handle **482**. Aside from minor visual differences, the main difference between coupler **410** and coupler **310** is that coupler **410** is configured to attach to a gooseneck trailer.

[0068] Coupler **510** shown in FIGS. **26-30** and coupler **511** shown in FIG. **31** are identical to each other, except for the bracketry used for mounting it to a trailer. Coupler **510** includes a mounting **512** for an “A frame” style trailer, while coupler **511** includes a flat mount bracket **514**. Couplers **510**, **511** also share many of the features of couplers **310** and **410** with certain differences noted below. Because coupler **511** is otherwise identical to coupler **510**, only one will be described in

detail.

[0069] As shown in FIGS. 27-28, the coupler 510 has a housing 540 with a fixed ball pocket 522. The housing 540 holds a pin 532 that is retained with a clip 533. The pin 532 holds a lever 530 that pivots between a captured, reset, and released position, and contains the same features as levers 330, 430. The lever 530 has the trip notch 336 and related surfaces as shown in FIG. 20. The lever 530 interfaces with a locking pin 560, shown in FIGS. 27-30. The locking pin 560 has a trip cam 376 and related surfaces shown as shown in FIG. 21A. The locking pin 560 has a major diameter 574, minor diameter 573, and a shoulder 575 therebetween. The locking pin 560 has helical grooves 564 that are different than helical grooves 364. Helical grooves 564 have an open end 561, end 568, and knee 572, but there is a radial portion 571 of the groove between the knee 572 and end 568 that is purely radial and does not turn back towards the end of the locking pin 560. Instead, it remains at a constant distance from the end of the locking pin 560. Screw in guides 590 each have a guide post 594 that remain in its corresponding groove 564 to guide the rotation of the locking pin 560. The engagement of the guide posts 594 to the grooves 564 is shown in FIG. 29. The locking pin 560 is held in a locking pin aperture 542, which allows movement of the locking pin 560 along and around a locking axis 562. A spring 570 is held between the shoulder 575 and a spring backstop surface 549 in the housing 540. As shown in FIG. 29, located between a release handle 582 and housing 540 is a seal or resilient spacer 550.

[0070] The trip cam 376, trip notch 336, and how the two cooperate is described in detail above, so only the differences specific to the locking pin 560 and lever 530 will be described. The locking pin 560 is moveable between a locked position, and armed position, and a released position through the operation of the release handle 582 or lever 530. In all of these positions, guide posts 594 remain in the grooves 564. In the released position, the guide posts 594 are in the radial portion 571. In the armed position, the guide posts 594 are between the knee 572 and the open end 561. The lever 530 pivots between a captured, released, and reset positions, described above.

[0071] Coupler 610 shown in FIGS. 32-36 has a different construction than couplers 510, 511 but otherwise operates using the same principles. The housing 640 is formed from two separate components, a formed or stamped housing portion 641 and an insert portion 643. The insert portion 643 is cast or machined from a solid piece of metal and retained to the stamped housing portion 641 with fasteners, welding, or a combination thereof. The housing 640 has a hemispherical surface 624 in a fixed ball pocket 622, shown in FIG. 36. The insert portion 643 has a locking pin aperture 642 that holds the locking pin 560. The insert portion 643 has a guard wall 644 that extends across the notch where the lever 630 is retained. The guard wall 644 connects sides of the insert portion 643 and prevents the hitch ball 12 from contacting the bottom of the lever 630 when the hitch ball is misaligned from the fixed ball pocket 622. The locking pin 560 and spring 670 are retained in the locking pin aperture 642 with the guide collar 390 having guide posts 394. The guide collar 390 is affixed to the housing 640 with fasteners. The guide posts 394 are located in the helical grooves 564 that guide the movement of the locking pin 560. The locking pin 560 interfaces with a lever 630 that has the trip notch 336. The lever 630 is retained by and pivots about a fastener 632 and a lock nut 633 that also retain the insert portion 643 to the stamped housing portion 641, shown in FIGS. 33 and 34. The fastener 632 extends through the insert and stamped housing portions 643, 641 as shown in FIGS. 33 and 34. A release handle 682 is affixed to the locking pin 560 with a seal 550 sandwiched between the guide collar 390 and release handle 682. The locking pin 560 passes through the spacer/seal 550, which acts as a resilient damper when the locking pin moves to the locked position.

[0072] For couplers 510, 511, and 610, the lever 530, 630 moves between a captured position, release position, and a reset position. As previously described, the captured position retains the hitch ball 12 while the release position allows the hitch ball 12 to enter or exit the fixed ball pocket 522, 622. The reset position of the lever 530, 630 is located between the captured and release positions and is defined by the angular position of the lever 530, 630 where the trip notch 336

contacts the trip cam **376**. Continued movement of the lever **530, 630** from the reset position to the release position rotates and moves the locking pin **560** from the released position to the armed position. When the guide posts **394** are located in the radial portion **571** (between the knee **572** and the end **568**), the locking pin **560** is in the release position. When the lever **530, 630** is moved from the reset position to the release position, the trip notch **336** engages the trip cam **376** to move the locking pin **560** from the release position. The guide posts **394, 594** are moved from the radial portion **571** over the knee **572**, which puts the locking pin **560** in the armed position and in biased contact with the side of the lever **530, 630**. When the hitch ball **12** is moved into the fixed ball pocket **522, 622**, the lever **530, 630** returns to the captured position, allowing the locking pin **560** to rapidly move to the locked position. The spring **570, 670** provides the biasing force on the locking pin **560**. The resilient spacer **550** acts as a cushion when the locking pin **560** moves from the released or armed position to the locked position. In the locked position, the spring **570, 670** maintains biased contact on the resilient spacer **550** between the release handle **582, 682** and housing.

[0073] A different self-resetting embodiment of the coupler **710** is shown in FIGS. **37-40E**. The coupler **710** shown can receive, couple to, and release the hitch ball **12**. As shown, the coupler **710** includes other features for a gooseneck application, but the same mechanism can be used with bumper pull or other ball-and-socket applications. The coupler **710** has a housing **740** with a fixed ball pocket **720** with a partially spherical inside surface **724**. The housing **740** also includes a slot **726** that extends into the fixed ball pocket **720** and a shaft aperture **742**. The shaft aperture **742** extends through the slot. The fixed ball pocket **720** has an opening **722** that is offset/shifted from the partially spherical inside surface **724**. As can be seen in section views FIGS. **40A-40E**, the opening **722** is offset from the fixed ball pocket and shifted towards a lever **730**. In other words, the hitch ball **12** must also translate horizontally with respect to the housing **740** once it passes through the opening **722** to fully seat within the fixed ball pocket **720**. Adjacent the opening **722** is a funneling surface **728** that is either conical or concave to direct the hitch ball **12** towards the opening **722**. The funneling surface **728** does not extend completely around the opening **722**. It is interrupted by a lever shroud **746** that flanks the slot **726** and protrudes below the funneling surface **728**. The lever shroud **746** protects the lever **730** from being damaged by a misaligned hitch ball during coupling.

[0074] The lever **730** is located in the slot **726** and is attached to a shaft **760**. The lever **730** includes a cam surface **732**, a reset portion **734**, a ball clearance cove portion **736**, and a ball engagement surface **738**. A keyed aperture **731** extends through the thickness of the lever and can receive a keyed end **764** on the shaft **760** to lock the lever **730** and shaft **760** together for rotation about a shaft axis **762**. The cam surface **732** is a curved/convex surface that has a center point that is offset from the shaft axis **762**. As will be described in greater detail below, the reset portion **734** is a protrusion that can contact the hitch ball **12** to reset the coupler. Located between the ball engagement surface **738** and reset portion **734** is the ball clearance cove portion **736** that is closer to the shaft axis **762** than the reset portion **734** and ball engagement surface **738**. The ball clearance cove portion **736** is concave and has a center point that is spaced from the shaft axis **762** by a first distance. The ball engagement surface **738** is concave and curved to match the outside diameter of the hitch ball **12**. The center point of the ball engagement surface is spaced from the shaft axis **762** by a second distance which is less than the first distance. A hold point **737** is located at the upper edge of the ball engagement surface **738**. The lever **730** also includes a release limit surface **733** and latch limit surface **735** that limit rotation of the lever by contacting a limit surface **744** in the housing.

[0075] The shaft **760** also has a bearing diameter **761** and a minor diameter **766** that allows the lever to rotate. At the end of the minor diameter **766** is a square that allows the shaft **760** to mate with a release handle **782**. A keyed bushing **750** has a bearing diameter **752** and a keyed portion **754** that is also received by the keyed aperture **731**. As can be seen in FIGS. **38-39**, a fastener **756**

secures the keyed bushing **750**, lever **730** and shaft **760**. The shaft aperture **742** and bearing diameters **752**, **761** allow the assembly to rotate. The minor diameter **766** is supported by a pin guide **790** that has a central aperture **792**. As can be seen in FIG. **39**, the pin guide **790** is secured to the housing **740** with a fastener.

[0076] A spring **770**, as shown in the FIGS, is formed from flat material into a “U” shape. A fixed portion **772** is in contact with an inside surface of the notch, and a moving portion **774** is connected to the fixed portion. The moving portion includes a bend **776** and a cam follower portion **778**.

[0077] The operation of the coupler **710** will now be described. As can be seen in FIGS. **40A-E**, the features on the lever **730** have interactions and relationships with other components, such as the hitch ball **12**, housing **740**, and spring **770**. The lever **730** is rotatable between a latched position (shown in FIGS. **40A** and **40C**) and a released position (shown in FIG. **40D**). The lever **730** rotates about the shaft axis **762** roughly **45** degrees between the latched and released position. The spring **770** remains in biased contact with the lever **730** throughout its rotation between the released position and latched position. In addition, the lever **730** has a counterweight portion **739** that protrudes beyond the lever shroud **746**. The counterweight portion **739** is located opposite the cam surface **732**. FIG. **40A** shows the lever **730** in the latched position and latch limit surface **735** contacting limit surface **744**. The hitch ball **12** has just started to enter the fixed ball pocket **720** where it is contacting the lever **730** and started to displace it from the latched position. As the hitch ball **12** continues towards the fixed ball pocket **720**, the lever **730** continues to rotate away from the latched position by the hitch ball **12** to the position shown in FIG. **40B**. In the positions shown in FIGS. **40A** and **40B**, the hitch ball **12** is adjacent to or located in the ball clearance cove portion **736**. As can be seen in FIGS. **40A** and **40B**, the cam follower portion **778** remains in biased contact with the cam surface **732**, urging the lever towards the latched position. FIG. **40B** represents the maximum rotation of the lever **730** as displaced by the hitch ball **12**, which is short of the released position. As the hitch ball **12** continues into the fixed ball pocket **720** (just before it contacts the inside surface **724**), the lever **730** is clear enough of the hitch ball **12** to return to the latched position. This is shown in FIG. **40C**, with the ball engagement surface **738** in biased contact with the outside spherical surface of the hitch ball. Any forces between the coupler **710** and hitch ball **12** only allow rotation between the two parts but not disengagement. To release the hitch ball **12** from the coupler **710**, the user rotates the release handle **782**, which in turn rotates the lever **730** to the released position shown in FIG. **40D**. In the released position, the hold point **737** contacts the spring **770**, lifting the cam follower portion **778** off of the cam surface **732**. In this released position, the lever **730** will remain in this angular position unless acted upon by another object, such as the release handle **782** or hitch ball **12**. As can be seen in FIG. **40D**, the ball clearance cove portion **736** faces the hitch ball **12** but the hitch ball cannot escape the fixed ball pocket **720** without contacting a portion of the lever **730**. This is shown in FIG. **40E**, where the distance between the opening **722** and the reset portion **734** is less than the outside diameter of the hitch ball **12**. As the hitch ball **12** continues to move away from the fixed ball pocket **720**, it contacts the reset portion **734**, moving the lever **730** away from the released position. Once the lever **730** is moved away from the released position, the cam follower portion **778** contacts the cam surface **732**, where the lever **730** is biased towards the latched position. As the hitch ball **12** continues to move away from the fixed ball pocket, the hitch ball **12** remains adjacent to the ball clearance cove portion and the lever returns to the latched position as shown in FIG. **40A**.

[0078] It is understood that while certain aspects of the disclosed subject matter have been shown and described, the disclosed subject matter is not limited thereto and encompasses various other embodiments and aspects. No specific limitation with respect to the specific embodiments disclosed herein is intended or should be inferred. Modifications may be made to the disclosed subject matter as set forth in the following claims.

Claims

1. A self-latching ball coupler for connecting to a hitch ball having a diameter, said ball coupler comprising: a housing having a fixed ball pocket with an opening, said housing having a slot adjoining and extending into said fixed ball pocket; a lever located in said slot, said lever having a ball engagement surface, a reset portion, and a ball clearance cove portion located between said ball engagement surface and said reset portion; a shaft fixed with respect to said lever, said shaft and said lever rotatable with respect to said housing between a latched position and a released position, in said latched position, said ball engagement surface faces said fixed ball pocket, in said released position, said ball clearance cove portion faces said fixed ball pocket; a release handle fixed with respect to said shaft and rotatable therewith; a spring in biased contact with said lever, when said lever is in said released position, said spring retains said lever in said released position, when said lever is moved away from said released position, said spring urges said lever towards said latched position; said hitch ball is removable from said fixed ball pocket when said lever is in said released position; when said hitch ball is in said fixed ball pocket and said lever is in said released position, when said hitch ball is removed, said hitch ball contacts said reset portion and moves said lever away from said released position.
2. The ball coupler of claim 1, wherein said housing has a limit surface facing said lever, said lever having a release limit surface adjacent a latch limit surface, said latch limit surface is obliquely angled with respect to said release limit surface.
3. The ball coupler of claim 1, said spring having a fixed portion in biased contact with said housing and a moving portion connected to said fixed portion, said moving portion having a follower portion, said lever having a cam surface, said follower portion spaced from said cam surface when said lever is in said released position, said follower portion is in biased contact with said cam surface when said lever is moved away from said released position.
4. The ball coupler of claim 3, wherein said spring is formed from a strip of flat material, said follower portion is obliquely angled with respect to said moving portion.
5. The ball coupler of claim 1, wherein said housing having a lever shroud portion flanking said slot and extending beyond said opening in a direction opposite said fixed ball pocket.
6. A self-latching ball coupler for coupling and releasing a hitch ball having a diameter, said ball coupler comprising: a housing having a fixed ball pocket complementary to said diameter, said fixed ball pocket having an opening, said housing having a slot adjoining and extending into said fixed ball pocket, said housing having a lever shroud portion flanking said slot and extending beyond said opening in a direction opposite said fixed ball pocket; a lever located in said slot and rotatable about a shaft axis between a latched position and a released position, said lever having a concave ball engagement surface spaced from said shaft axis by a first distance, said concave ball engagement surface extending to a concave ball clearance cove surface spaced from said shaft axis by a second distance, said concave ball clearance cove surface terminating at a reset portion, in said latched position, said concave ball engagement surface faces said fixed ball pocket, in said released position, said concave ball clearance cove surface faces said fixed ball pocket; a release handle fixed with respect to said lever and rotatable therewith; a spring in biased contact with said lever, when said lever is in said released position, only said spring retains said lever in said released position, when said lever is moved away from said released position, said spring urges said lever towards said latched position; said hitch ball is removable from said fixed ball pocket when said lever is in said released position; when said lever is in said released position, said reset portion extends into said opening to narrow said opening smaller than said diameter; and when said hitch ball is removed from said fixed ball pocket, said hitch ball contacts said reset portion and moves said lever away from said released position.
7. The ball coupler of claim 6, said spring having a fixed portion in biased contact with said

- housing and a moving portion connected to said fixed portion, said moving portion having a follower portion, said lever having a cam surface, said follower portion spaced from said cam surface when said lever is in said released position, said follower portion is in biased contact with said cam surface when said lever is moved away from said released position.
- 8.** The ball coupler of claim 7, when said lever is in said latched position and said hitch ball is moved into said fixed ball pocket, said lever remains in biased contact with said hitch ball and said follower portion remains in contact with said cam surface.
- 9.** The ball coupler of claim 7, wherein said spring is formed from a strip of flat material, said follower portion is obliquely angled with respect to said moving portion.
- 10.** The ball coupler of claim 6, wherein said first distance is greater than said second distance.
- 11.** The ball coupler of claim 6, wherein said lever has a latch limit surface and a release limit surface, when said lever is in said released position, said release limit surface is adjacent said housing, when said lever is in said latched position, said latch limit surface is adjacent said housing.
- 12.** The ball coupler of claim 6, wherein said lever protrudes beyond said lever shroud portion.
- 13.** The ball coupler of claim 6, wherein removing said hitch ball from said fixed ball pocket resets said lever to said latched position.
- 14.** A self-latching ball coupler for connecting to a hitch ball having a diameter, said ball coupler comprising: a housing having a fixed ball pocket with an opening, said housing having a slot adjoining and extending into said fixed ball pocket; a lever located in said slot and rotatable about a shaft axis between a latched position and a released position, said lever having a first concave surface extending to a second concave surface, said second concave surface terminating at a reset portion, said lever having a convex surface spaced from said first and second concave surfaces, in said latched position, said first concave surface faces said fixed ball pocket, in said released position, said second concave surface faces said fixed ball pocket; a release handle fixed with respect to said lever and rotatable therewith about said shaft axis; a spring in biased contact with said lever, when said lever is in said released position, said spring retains said lever in said released position, when said lever is moved away from said released position, said spring contacts said convex surface to urge said lever towards said latched position; and when said hitch ball is in said fixed ball pocket and said lever is in said released position, said reset portion partially extends into said opening to narrow said opening and when said hitch ball is removed, said hitch ball contacts said reset portion and moves said lever away from said released position.
- 15.** The ball coupler of claim 14, when said hitch ball passes through said opening and into said fixed ball pocket, said lever approaches but does not reach said released position.
- 16.** The ball coupler of claim 14, wherein removal of said hitch ball from said ball coupler resets said lever to said latched position.
- 17.** The ball coupler of claim 14, said spring having a fixed portion in biased contact with said housing and a moving portion connected to said fixed portion, said moving portion having a follower portion, said follower portion spaced from said convex surface when said lever is in said released position, said follower portion is in biased contact with said convex surface when said lever is moved away from said released position.
- 18.** The ball coupler of claim 17, when said lever is in said latched position and said hitch ball is moved into said fixed ball pocket, said lever remains in biased contact with said hitch ball and said follower portion remains in contact with said convex surface.
- 19.** The ball coupler of claim 17, wherein said spring is formed from a strip of flat material, said follower portion is obliquely angled with respect to said moving portion.
- 20.** The ball coupler of claim 14, said housing having a lever shroud portion flanking said slot and extending beyond said opening in a direction opposite said fixed ball pocket.
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